Assignment 3

Fumiya Nagatomo

March 3, 2026

1 Introduction

This assignment explores the concept of Singular Value Decomposition (SVD) and its application in block-wise image compression. The goal is to analyze the trade-off between compression ratio and image quality by reconstructing images using the top-k singular values.

2 Methodology

The grayscale image is divided into non-overlapping 8×8 blocks. For each block, Singular Value Decomposition is applied:

$$A = U \Sigma V^T$$

The image is reconstructed using only the top-k singular values:

$$A_k = U_k \Sigma_k V_k^T$$

where $k = 1, 2, \ldots, 8$.

3 Implementation

The algorithm consists of the following steps:

- Load image and convert grayscale
- Divide image into 8×8 blocks
- Apply SVD to each block
- Reconstruct blocks using top-k singular values
- Compute compression ratio and reconstruction error

4 Results

As k increases, more singular values are retained. This improves image quality but reduces compression efficiency.

The compression ratio decreases as k increases, while reconstruction error improves.

5 Discussion

The experiment demonstrates the trade-off between compression ratio and reconstruction error. Smaller k values achieve higher compression but lower image quality. Larger k values preserve more details but reduce compression ratio.

6 Reconstructed Images

Here are the reconstructed images for 8×8 blocks

$k = 1$

$k = 2$

$k = 3$

$k = 4$

$k = 5$

$k = 6$

$k = 7$

$k = 8$